

MicroACT Repo Reading Order

A recommended path for someone new who wants to read every source file and understand how the whole codebase fits together.

High-Level Path

README -> config -> dataset -> training loop -> policy wrapper -> CVAE -> transformer/backbone -> visualization/export tools.

Order	File	Why Read It Here
1	README.md	Read fully first. It explains ACT, the dataset shape, training, inference, and the intended mental model.
2	requirements.txt	Establishes the runtime world: PyTorch, torchvision, numpy/pandas/PIL, and optional Cellpose.
3	.gitignore	Clarifies what is source versus generated artifacts: checkpoints, ONNX exports, torchviz exports, dataset frames/videos.
4	config/config.py	The single source of truth for paths, state/action dimensions, image size, ACT hyperparameters, training defaults, and rollout placeholders.
5	data/dataset.py	Shows how raw demonstrations become tensors: CSV loading, image-path fallback, normalization stats, action chunk padding, and <code>is_pad</code> .
6	utils.py	Small but important training support: seeding, AdamW parameter groups, checkpoint save/load, and metric averaging.
7	train.py	The main execution flow: parse CLI, build dataset, split train/val, build policy, optimizer, epoch loop, checkpointing.
8	model/act_policy.py	The practical model interface. Wraps the CVAE, computes masked L1 + KL loss, stores normalization buffers, and exposes numpy inference.
9	model/cvae.py	The heart of the model: style encoder, latent sampling, image encoding, qpos/latent tokens, transformer decoder, and action head.
10	model/transformer.py	The DETR-style encoder/decoder primitives used by the CVAE. Explains sequence-first tensors and position injection.
11	model/backbone.py	Read after <code>cvae.py</code> so the return shapes matter. Covers ResNet18, DINOv2, Cellpose, dual encoders, projections, and 2D sine positions.
12	viz_summary.py	Side tool for layer, shape, and parameter summaries. Useful once the model code is familiar.
13	export_onnx.py	Side tool for exporting inference and training graphs to Netron. Shows the two model paths cleanly.
14	viz_torchviz.py	Side tool for autograd graph SVGs. Best read last among scripts because it assumes you already know the model pieces.

Order	File	Why Read It Here
15	<code>evaluate.py;</code> <code>config/__init__.py;</code> <code>data/__init__.py;</code> <code>model/__init__.py</code>	These are empty. Note their package or placeholder role.
16	<code>dataset/;</code> <code>checkpoints/;</code> <code>onnx_exports/;</code> <code>torchviz_exports/;</code> <code>__pycache__/;</code> <code>.claude/;</code> <code>.codex</code>	Read conceptually, not as source. These are inputs, generated outputs, Python caches, or local tool metadata.

Source Surface Checked

README.md, requirements.txt, .gitignore, config/config.py, data/dataset.py, utils.py, train.py, model/act_policy.py, model/cvae.py, model/transformer.py, model/backbone.py, viz_summary.py, export_onnx.py, viz_torchviz.py, evaluate.py, and empty package __init__.py files.

The code also passed: `python3 -m py_compile config/config.py data/dataset.py model/backbone.py model/transformer.py model/cvae.py model/act_policy.py utils.py train.py export_onnx.py viz_summary.py viz_torchviz.py evaluate.py`